

The Charge Checker: How It Works and How to Use It

A plain-English guide for the Operations Director | KPC Revenue Assurance | Brian Mugambi

1. The Analogy: A Panel of Experienced Billing Auditors

Imagine a panel of 300 auditors who have each studied thousands of past gate records and learned what a suspicious waiting-fee charge looks like. When a new charge arrives, every auditor looks at it from a slightly different angle and votes. If enough of them vote "take a second look", the charge is put on a short list for a human reviewer. **The Charge Checker is that panel, in software.** It does not accuse anyone and it never bills, cancels or changes a charge. It only decides which charges deserve your team's attention first.

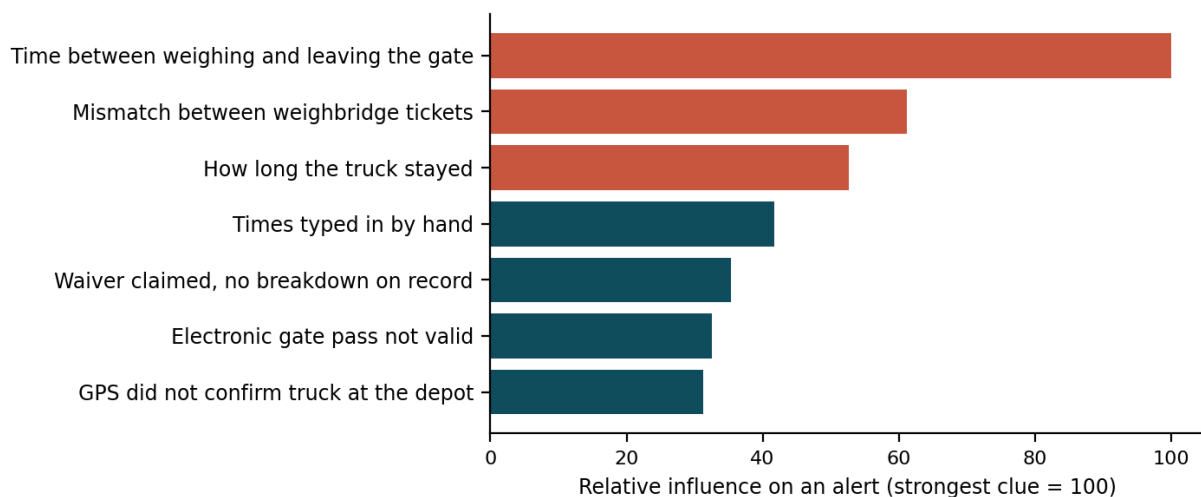
2. "Isn't It Just Guessing?" The Evidence

A fair question. We tested the checker on **3,000 charges from months it had never seen**, where we knew the real answers. Of those, 52 were genuinely problematic.

- It flagged 170 charges (under 6% of the total) and **43 of them were real problems, so it caught about 8 in every 10 (43 of 52).**
- Picking 170 charges at random would have found only about **3** of the 52.
- A checker that flags nothing would be "98% accurate", because 98% of charges are fine, yet it would catch none. That is why we judge the tool on problems caught and false alarms, not on overall accuracy.

Important: this test used **simulated data** built to resemble KPC gate records, because the platform does not yet hold enough real history to train on. The results show the method works; a live pilot must confirm the real figures.

3. What It Looks At: The Three Biggest Clues



Longer bar = bigger influence on whether a charge is flagged. The three strongest clues are highlighted.

- **Time between weighing and leaving the gate.** In our test data a truck normally leaves about half an hour after weighing. When the recorded gap is hours long, or the times do not add up, the checker treats it as a strong warning that the timestamps may be wrong.
- **Mismatch between weighbridge tickets.** The weighbridge issues two tickets for each truck, and their records are expected to agree within a small margin. When they do not, something may be wrong with a ticket or the load record.
- **How long the truck stayed.** Very long stays raise the risk, especially together with the other clues. On its own, a long stay is often legitimate; the checker looks at how the clues combine.

A Real Alert, Explained

Charge flagged with a very high risk score. Reasons shown to the reviewer, in order: (1) the truck left the gate **about 5 hours after weighing**, against a normal half hour; (2) the gate times were **typed in by hand**; (3) the stay was **19.5 hours**. The reviewer checks these against the gate log and decides. Every alert comes with its reasons, so nobody has to take the tool's word for it.

4. Trust and Limits: Where It Gets Things Wrong

No checker is perfect, and you should know exactly how this one fails.

- **False alarms are expected, and they are deliberate.** About **3 in 4 flagged charges will turn out to be fine** (127 of the 170 in our test). Think of a smoke detector that goes off when someone burns the toast: irritating, but far better than missing a real fire. We set the sensitivity high on purpose, because missing a problem charge costs far more than a few minutes of review.
- **Some real problems will be missed.** About 1 in 6 problem charges slipped through in our test (9 of 52). A charge that is not flagged is **not proven fine**.
- **Small test, wide margin.** With only 52 problem cases, the true catch rate could plausibly sit anywhere from about 7 in 10 to 9 in 10.
- **It only knows what it has seen.** A new type of irregularity, a change in gate procedure, or a faulty sensor can confuse it. Reviewer outcomes should be recorded monthly so the tool can be re-checked and retrained.

How the Operations Team Should Use It

Treat the Charge Checker as a guide, not a replacement for judgment. Use it to sort the day's charges so reviewers start with the riskiest. Read the reasons shown, check them against the gate and weighbridge records, and make the call yourself. If your experience says an alert is wrong, trust your experience, and tell us, because that feedback makes the tool better.

At a Glance: The Test Results

What we measured	Result	In plain terms
Charges tested (months it had not seen)	3,000	52 were genuinely problematic
Problem charges caught	43 of 52	About 8 in 10
Problem charges missed	9 of 52	About 1 in 6
Charges put on the review list	170	Under 6% of all charges
Flagged charges that were fine	127 of 170	About 3 in 4 (the smoke-detector effect)

Simulated data. Figures are for illustration of the method and must be confirmed in a live pilot.

Proposed Next Steps

- Run a **one-depot pilot for one full billing cycle**, with reviewers recording whether each alert was a real problem.
- Meet with Operations and Finance to **agree the sensitivity**: more alerts means fewer misses but more review work.
- Re-check the tool **every month** against reviewer outcomes, and retrain it as real gate history builds up.